

Tutorial Session

9 novembre 2023

Exercices

The "DEMO" data sheet

Calculate the potential impact on the classification quality indicator of an association between Germany and Bulgaria. Comment on this calculation, specifying the relative importance of the different variables. Can you propose a solution to correct this issue ?

Perform the two applications of agglomerative hierarchical clustering with the raw data, and then apply your solution to correct the previously mentioned problem. Comment on the two sets of results obtained.

The k-means approach

Compare the results obtained for the same number of groups using the agglomerative hierarchical clustering technique and the technique called 'k-means.' The instruction to proceed is as follows : `'kmeans(data, centers = nb_clusters)'`. Please replace `'nb_clusters'` with the actual number of clusters you intend to use in your analysis.

Case Study : Conglomerate Inc's New PDA Case

The cellular phone division of Conglomerate Inc. has teamed up with a PC manufacturer to develop, produce, and market a novel hybrid combination of a personal digital assistant (PDA) and a "smart" cellular phone. They have tentatively named it ConneCtor. It transmits and receives both data and voice (unlike competing PDAs which focus on data).

ConneCtor Another Conglomerate Success Story ?

ConneCtor is lightweight and is shaped like a portable phone with a small backlit LCD touch screen along the handset. Its (open) operating system performs standard cellphone functions and such personal information management (PIM) functions as a calendar, calculator, and address book. It can send and receive faxes, voice messages, and email. Users can input data in four ways :

- By typing on the screen keyboard



FIGURE 1 – Personal Digital Assistants

- By using a numerical keyboard
- By writing on the screen in "digital ink"
- By speaking into the phone (it includes voice recognition software)

The voice recognition feature is based on a neural network that is trained to recognize a particular user's voice patterns. An additional feature unique to ConneCtor is linkage via wireless local area networks to other PDA's. In summary, the features of Conglomerate's handheld device are :

- Instant communication from PDA to PDA
- Cellular phone and pager, fax and e-mail
- Calendar, scheduler, calculator, and address book
- An open system for customized applications
- A paperless note pad
- Voice recognition

Conglomerate is now trying to identify segments within the market for PDAs, target appropriate segment(s) for ConneCtor, and position ConneCtor in the chosen segments.

Background on the PDA market

In August 1993, Apple introduced its Newton PDA. The broad acceptance that Apple had anticipated did not materialize, and Apple sold only 80,000 Newtons that year. In 1995, the PDA appeared to be on the verge of greater growth and development. The PDA market had grown in four areas : specialized vertical applications (e.g., physician scheduling), PIM (Personal Information Management), mobile communications, and as a supplementary gateway into the Internet. Even though it has four main applications, the PDA is primarily targeted at "road warriors" or "mobile professionals." This group consists of approximately 25 million people in the United States, of whom about 5 million travel with a computer notebook. Many of these individuals already have cellular phones and must send and receive a large number of messages and data. The standard PDA cannot handle their needs.

The survey

Conglomerate, Inc. hired a market research firm to survey the market across a broad range of occupation types. The survey includes a screening item asking respondents if they had or

would consider a PDA. Only those respondents who answered affirmatively to that question were retained for further analysis.

The questionnaire

The questionnaire asked the respondents to provide data on two kinds of variables : segmentation basis or needs variables and variables that could be used in describing or targeting the clusters.

Questions for determining segmentation

- X1 Whenever new technologies emerge in my field, I am among the first to adopt them. (1 = Strongly disagree 7 = Strongly agree)

How often do you use the following : (1 = Never 7 = Always)

- X2 a. Pager ?
- X3 b. Phone or voice mail ?
- X4 c. Scheduling or contact-management tools, i.e., filofax or similar devices ?
- X5 How often do others send you time-sensitive information (e.g., work orders)? (1 = Never 7 = Daily)
- X6 How often do you have to send time-sensitive information while away from your office? (1 = Never 7 = Daily)
- X7 How much of your time do you spend away from your office location? (1 = 0 % 7 = 70% or more)
- X8 How important is wireless communication to you? (1 = Not at all important 7 = Very important)
- X9 How important is it for you to share information rapidly with colleagues while away from an office location? (1 = Not at all important 7 = Very important)

How much would you be willing to pay for a personal digital assistant (PDA) with the following features : instant communication from PDA to PDA, cellular phone and pager, fax and email, calendar, scheduler, calculator, address book, open system for customized applications, paperless note pad, and voice recognition ?

- X10 a. Monthly (for all services that you use) ?
- X11 b. Invoice price for the PDA device with all features ?

Questions for describing or targeting the clusters

- Z1 Age
- Z2 Education (1 = High school, 2 = Some college, 3 = College, 4 = Graduate degree)
- Z3 Income Type of industry or occupation : (0 = No, 1 = Yes)
- Z4 Construction
- Z5 Emergency (fire, police, ambulance, etc.)
- Z6 Sales (insurance, pharmacy, etc.)
- Z7 Maintenance and service
- Z8 Professional (e.g., lawyer, consultant, etc.)
- Z9 Do you own a PDA ? Media consumption (Readership of magazines) : (0 = No, 1 = Yes)
- Z10 Businessweek
- Z11 PC Magazine
- Z12 Field & Stream
- Z13 Modern Gourmet

Questions

1. Run only cluster analysis on the data to try to identify the number of distinct segments present in this market. Consider both the distances separating the segments and the characteristics of the resulting segments.
2. Identify and profile (name) the clusters that you select. Given the attributes of ConneCtor, which cluster would you target for your marketing campaign?
3. Describe the segments using the answers of the second set of questions. How would you go about targeting the segment(s) you picked in question 2?
4. How has this analysis helped you to segment the market for ConneCtor?